JOB MARKET ANALYSIS REPORT

Project PRDA-04 | Data Science Job Market, analysed in Microsoft Power BI

Introduction

The objective of this project is to analyse the Data Science job market and identify trends in job openings, salaries, industries, companies, job titles, education and technical skills.

The dataset contains 742 job postings and 42 columns, including Job ID, Job Title, Company Name, Location, Industry, Degree, Lower Salary, Upper Salary, Average Salary and a set of skill columns. The purpose of the project is to show where the jobs are, what they pay and which skills employers ask for, so that a company can plan its hiring with this information.

The analysis was performed using Microsoft Power BI. Three interactive dashboards were created: Job Market Overview, Salary Analysis and Skills Analysis. Each dashboard has filters for industry, location, job title and other fields, so the numbers can be viewed for a chosen group.

Data and Measures

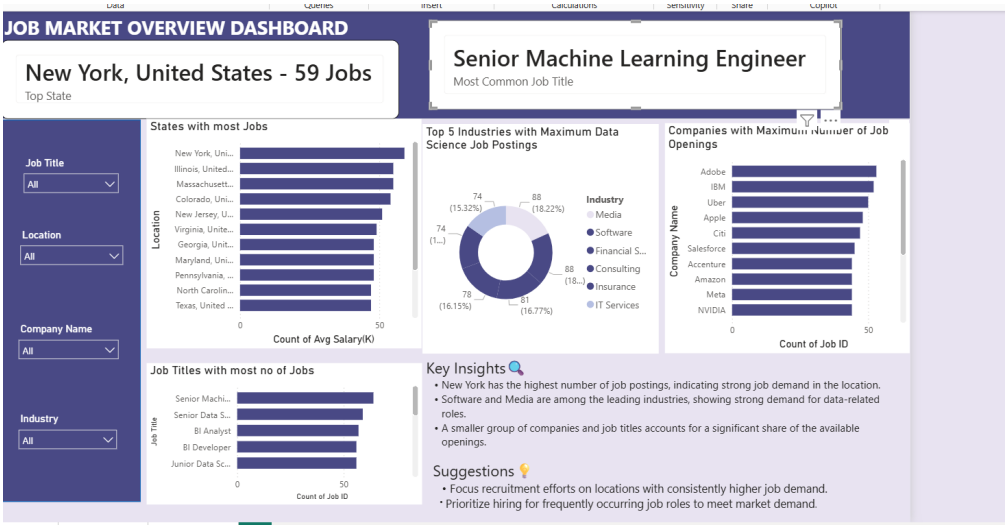
The main table contains one row for each job posting, identified by its Job ID. Total job postings are calculated as the count of Job ID.

The dataset contains salary fields including Lower Salary, Upper Salary and Avg Salary(K). The average salary is expressed in thousands of US dollars.

The skills are stored in a separate Skills table linked to the main table. It has a Skills column and a Required column, which makes it possible to show the skills needed for each job title.

Measures were created for the dashboard cards: Top State, Top Company, Most Common Job Title, Highest Average Salary State, Highest Average Salary Job Title and Most Requested Skill.

Dashboard 1: Job Market Overview



The first dashboard analyses job opportunities across different states, industries, companies and job titles. It has cards for the top state (New York, 59 jobs) and the most common job title, and filters for job title, location, company name and industry.

New York has the most job postings, followed by Illinois and Massachusetts. The bars for the other states come down gradually, which shows that no single state dominates the market.

The industry chart shows that Software and Media are among the leading industries, and that Financial Services, Consulting, Insurance and IT Services have similar posting counts. This shows that Data Science roles are available across several industries.

The company chart shows that Adobe has the most job openings, followed by IBM and Uber. The job title chart shows that Senior Machine Learning Engineer is the most common job title, followed by Senior Data Scientist, BI Analyst and BI Developer.

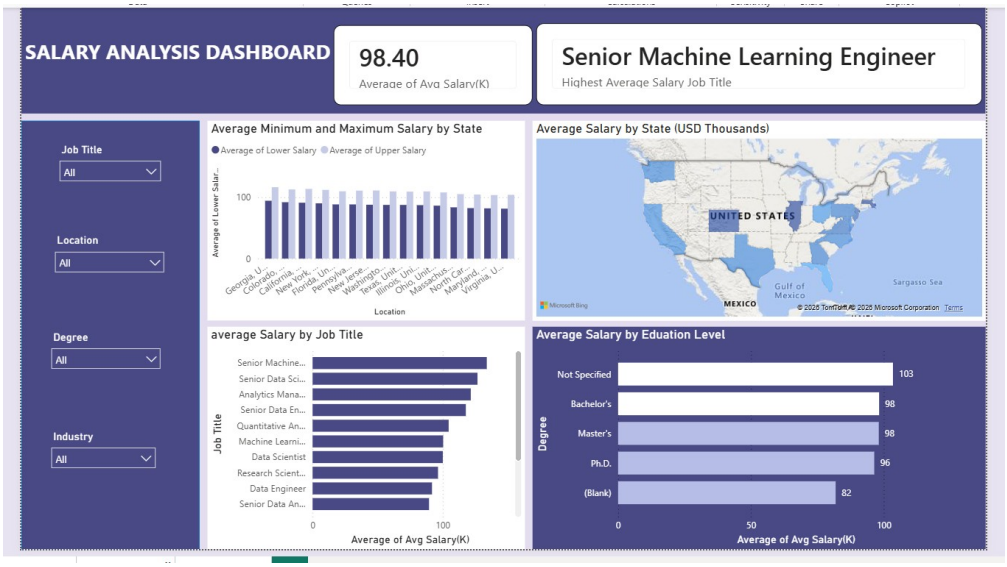
Key Insights

- New York has the most job postings, which points to strong demand in that location.
- Software and Media are among the leading industries hiring for data-related roles.
- Adobe, IBM and Uber are the companies with the most openings.
- Senior Machine Learning Engineer is the most common job title.

Recommendations

- Companies should focus recruitment efforts on locations with consistently higher job demand.
- Hiring should be prioritised for frequently occurring job roles to meet market demand.

Dashboard 2: Salary Analysis



The second dashboard analyses salary differences across states, job titles and education levels. It has cards for the overall average salary (98.40 thousand) and the highest average salary job title, a map of average salary by state, and filters for job title, location, degree and industry.

Georgia has the highest average salary, while New York has the most job postings. This shows that the state with the most jobs is not necessarily the state that pays the most. The comparison of average

minimum and maximum salaries shows that salaries vary across states, although the differences between states are small.

The salary comparison by job title shows that Senior Machine Learning Engineer, Senior Data Scientist and Analytics Manager are among the highest paid roles, clearly above roles such as Data Scientist and Research Scientist.

The map shades each state by its average salary, so darker states pay more. The education bar chart compares the average salary of the groups Not Specified, Bachelor's, Master's, Ph.D. and Blank. The averages range from about 82 to about 103 thousand, so salary differs by education level.

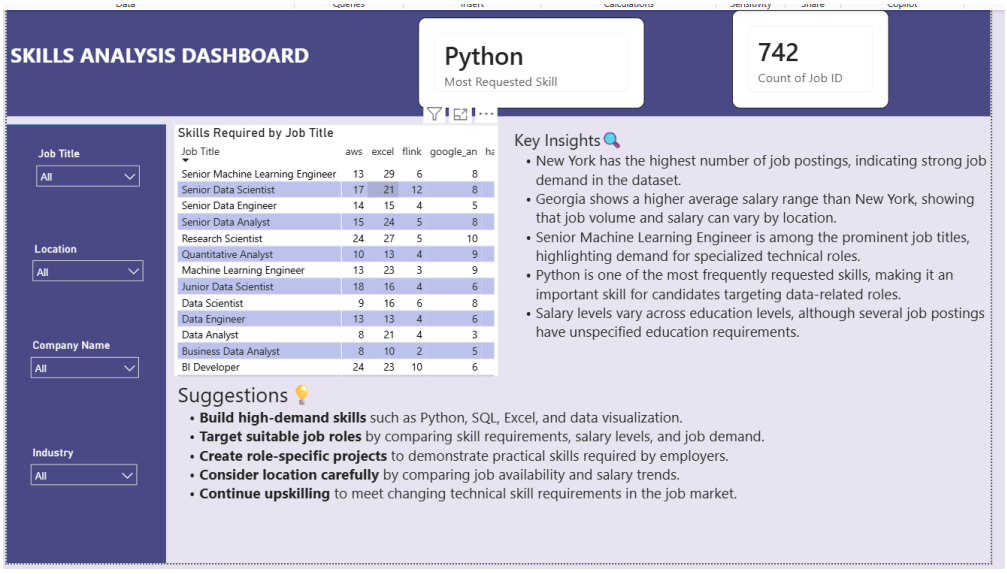
Key Insights

- Georgia has the highest average salary, not New York.
- Average minimum and maximum salaries vary by state, but only slightly.
- Senior roles are among the highest paid.
- Average salary differs by education level.

Recommendations

- Salary offers should be set by job title, seniority, location and market demand and not by one average figure.
- Salary ranges should be reviewed regularly to stay in line with the market.

Dashboard 3: Skills Analysis



The third dashboard focuses on the skills required for different job titles. It has cards for the most requested skill and the total number of job postings (742), and a table that shows how many times each skill is required for each job title. It can be filtered by job title, location, company and industry.

Python is the most requested skill. The skills table shows that the requirements change with the job title. For example, Research Scientist postings ask for AWS 24 times and Excel 27 times, while Senior Machine Learning Engineer postings ask for AWS 13 times and Excel 29 times.

Key Insights

- Python is the most requested skill.
- Skill requirements differ by job title.
- Salary levels also vary across job titles and education levels.
- Looking at salary and skills together shows which skills are in demand and where pay is competitive.

Recommendations

- Candidates with in-demand technical skills such as Python, SQL, Excel and data visualization should be prioritised.
- Role-specific training and upskilling programmes can help address skill gaps, and skill requirements should be reviewed regularly as the market changes.

Overall Findings and Conclusion

The analysis shows that the Data Science job market depends on several factors together, including location, industry, company, job title, skills, education and salary.

The major findings of the analysis are:

- New York has the most job postings, but Georgia has the highest average salary.
- Data Science roles are available across several industries, with Software and Media among the leaders.
- A few companies, led by Adobe, IBM and Uber, have the most openings.
- Senior Machine Learning Engineer is the most common job title and is also among the highest paid.
- Salary varies across states, job titles and education levels.
- Python is the most requested skill, and other skill requirements differ by role.

Final Recommendations

- Focus recruitment on locations with consistently higher job demand.
- Prioritise hiring for frequently occurring roles such as Senior Machine Learning Engineer and Senior Data Scientist.
- Offer competitive salaries based on job role, location, skills and market demand.
- Prioritise candidates with in-demand skills such as Python, SQL, Excel and data visualization.
- Provide role-specific training and upskilling to address skill gaps.
- Review salary ranges and skill requirements regularly to stay aligned with the market.
- Use the salary and skill data together when planning hiring, training and workforce needs.